From Equity Pairs to Options Relative Value

Does reinforcement learning improve capital allocation?

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Abstract

We extend equity-pairs trading to relative implied volatility in approximately 30-day, 25-delta out-of-the-money puts on related equity underlyings. A screened opportunity rule generates trades; direct policy-search reinforcement learning (RL) then determines their entry size. In a simulated $10 million account over January 2022–August 2025, constant full sizing achieves annualized excess Sharpe 1.055, compared with 1.263 for decision-tree states and cross-entropy sizing. The observed gain of 0.208 has a conditional 95% interval of $[-0.103, 0.424]$. The learner admits 110 trades compared with 75 under constant full sizing. Performance varies across years, and previously inspected history, different training and evaluation conventions, and simplified execution limit the conclusions. The comparison supports positive historical profit above cash and a higher point estimate for learned sizing, without establishing a repeatable advantage from learning.

1 Introduction

Suppose AMD and NVDA face similar news, but AMD options become unusually expensive relative to NVDA options. Selling the richer options and buying the cheaper ones expresses a view about the relative price of volatility. Daily stock hedges reduce immediate directional exposure, and initial vega matching reduces sensitivity to a common small change in implied volatility. The trade still carries risk: related stocks can experience different events, and option premiums may correctly reflect those differences.

Our starting point is Ning and Lee's equity-pairs framework [1], which separates constructing a spread from learning how to trade it. We carry that distinction into options. First, we ask whether a screened relative-volatility rule produces useful historical trades under explicit contract and funding assumptions. We then ask the central question: *does learning improve how capital is allocated to those opportunities?* A profitable signal and a useful allocation policy are separate findings.

After explaining the motivating paper and our adaptation, we define the trade, monthly training procedure and tree/CEM sizing method. We then compare constant and learned sizing in the same funded account and examine the learner's annual results. The comparison uses the retained configuration and saved historical backtests.

2 Motivating paper and adaptation

What the paper does

Ning and Lee [1] study how to construct and trade mean-reverting equity spreads. At time t, let $S_{1,t}$ and $S_{2,t}$ denote the two stock prices. They form the spread $X_t = S_{1,t} - BS_{2,t}$, where B is the hedge coefficient, and choose B by grid search to minimize empirical mean reversion time: the average duration from a significant local extreme to the next mean crossing. Formation does not fit an Ornstein–Uhlenbeck (OU) mean-reversion model.

For trading, they use tabular Q-learning. The state encodes recent spread moves by direction and magnitude; available actions buy, hold or sell a unit, subject to the current position. The policy enters a long spread and later closes it, and does not initiate a short spread. Its reward favors buying below and selling above the equilibrium mean, with a transaction-cost term $c|A|$, where c is the cost coefficient and $|A|$ is the magnitude of the trading action. Q-values combine immediate reward with the estimated value of later decisions.

Because the reward requires a true equilibrium mean, the agent trains on 10,000 one-year simulated OU paths before application to historical stocks. The market study forms ten US pairs in 2022 and trades them in 2023. Published daily Sharpe ratios exceed the Distance Method and OU benchmarks for all ten pairs. Those daily figures are not directly comparable with our annualized options Sharpe ratios.

Criticisms that motivate our design

Three issues limit interpretation. **Short history:** one trading year provides limited evidence across regimes. **Costs and scale:** the reward defines a cost term, but its numerical implementation and reconciliation to net performance are not clearly documented. This is not evidence that costs were omitted. Higher turnover raises total costs; larger or faster orders can also increase per-unit slippage and market impact. A constant coefficient does not establish scalability. **Simulation transfer:** formation is empirical, but policy training still uses specified mean-reverting dynamics and a known equilibrium. Real relationships can shift, so extensive synthetic experience does not establish market transfer.

What we borrow and what we change

We borrow the separation of opportunity construction from trading decisions, and the question of whether learning improves on a simple rule. We change the instrument from equity-price spreads to option-implied-volatility spreads, using actual contracts, initial vega matching and daily stock delta hedges. Completed historical outcomes replace a known synthetic equilibrium for training. We make simulated costs explicit and compare multiple trading years.

The next section defines relative-IV opportunities, option contracts and hedging through expiry. Our regression tree groups completed historical opportunities; direct policy-search RL chooses an entry size for each group using the cross-entropy method (CEM). Comparing with constant sizing measures the additional historical value of learned allocation.

3 Our strategy and evaluation

3.1 From a volatility signal to a hedged trade

Implied volatility (IV) reproduces an option quote under a pricing model; realized volatility (RV) measures stock-price variation. RV inputs use trailing data available at the signal date. We trade puts near 25 delta and 30 days to expiry (P25), retained after exploratory research and fixed across sizing policies. We do not claim that this segment is optimal. These puts have strikes below the stock price. At delta -0.25, a long 100-share contract initially has approximately -25 shares of stock exposure.

For underlyings A and B, let $IV_{A,t}$ and $IV_{B,t}$ be selected-option IVs on signal date t. Rank their difference $D_t = IV_{A,t} - IV_{B,t}$ within the 252-session history ending on that date, with at least 63 observations. Above the 80th percentile, sell A's relatively rich put and buy B's; at or below the 20th percentile, reverse the direction. The percentile compares the pair with its own history, allowing for persistent IV differences.

At monthly formation, estimate beta of A's returns on B's and correlation using the preceding 126 sessions. Require beta between 0.85 and 1.15 and correlation of at least 0.5. Also require positive average net profit per initial vega for the proposed direction in full-size standalone simulations completed within the preceding 91 and 365 calendar days, with at least 21 and 63 outcomes, respectively. These simulations use training costs, can overlap and need not have been executed by the funded account. An unfinished trade's eventual profit cannot inform a current decision.

Quantities approximately match initial vega: option-value sensitivity to a one-percentage-point IV increase. Table 1 illustrates the entry positions at delta -0.25 per share and 100 shares per contract.

Position	Contracts	Vega per IV point	Initial stock hedge
Buy A puts	+100	+\$450	Buy 2,500 shares
Sell B puts	−150	−\$450	Short 3,750 shares

Table 1: Illustrative vega matching: A contracts have \$4.50 vega each; B contracts have \$3.00.

Hold each option to expiry and adjust stock delta hedges daily. Profit combines premiums, settlement payoffs and realized hedge cash flows, after costs and financing. The IV-rank signal is a hypothesis about relative option richness; initial vega matching and temporary IV convergence do not guarantee profitable expiry outcomes.

3.2 Simulating trades in a funded portfolio

Each policy uses one account initially funded with $10 million. Historical quotes, prices, rates and dividends come through WRDS. We inherit an earlier 345-pair roster and formation calendar; 125 pairs with 161 equity and ETF underlyings supply prepared candidates. Of these, 46 pairs generate 608 screened proposals over 2018–2025, including 312 in the later period. The universe is conditioned on prior research and available data, rather than independently selected for this comparison.

Contracts are selected on the signal date and traded, if eligible, on the next session. Whole-contract quantities must leave at most 10% initial vega mismatch. Accepted positions share cash and capital; stock hedges in the same name are netted. Existing trades can therefore prevent new entries. All policies receive identical proposals, quote checks and admission rules; Appendix A defines the contract and risk checks.

Option trades fill at midpoint with an execution charge equivalent to 0.10 IV percentage point of traded vega. For trading day t, let $\mathcal{F}_t$ be paid option fills, $\Delta n_{j,t}$ contracts traded in option j, and $V_{j,t}$ one contract's dollar vega per IV point. For stock s, let $P_{s,t}$ be price and $\Delta h_{s,t}$ the change in net hedge shares. Dollar execution cost is

$$C_t^{\text{exec}} = \sum_{j \in \mathcal{F}_t} |\Delta n_{j,t}|(0.10 V_{j,t} + 0.03) + 0.0001 \sum_s P_{s,t} |\Delta h_{s,t}|.$$

The constants are $0.03 per option contract and one basis point of stock turnover. Expiry settlement carries no option execution charge.

Let B_{t-1} be prior cash, H_{t-1}^- short-stock notional and $\Delta\tau_t$ elapsed calendar days divided by 365. Dollar funding charges are

$$C_t^{\text{fund}} = [0.003 H_{t-1}^- + 0.01 \max(-B_{t-1}, 0)]\Delta\tau_t.$$

These charge 0.3% annual short-stock borrow and a 1% surcharge on negative cash. Cash interest is $I_t = r_{t-1} B_{t-1} \Delta\tau_t$, where r_{t-1} is the prior annual benchmark rate as a decimal. Negative cash pays both that rate and the surcharge. Corporate-event exits, missing-quote handling and exercise approximations are summarized in Appendix A.

Because unused cash earns interest, we evaluate trading value above an all-cash benchmark, including premiums, settlement, hedges, dividends, costs and financing. Total account growth alone does not measure trading performance.

3.3 Training, selection and testing

Each month, we train a new tree and size map using historical examples from the preceding 18 months, then apply that model to the next month's entry decisions. Figure 1 separates this rolling process from the choice of model specification. Training and application use successive slices of the same market history.

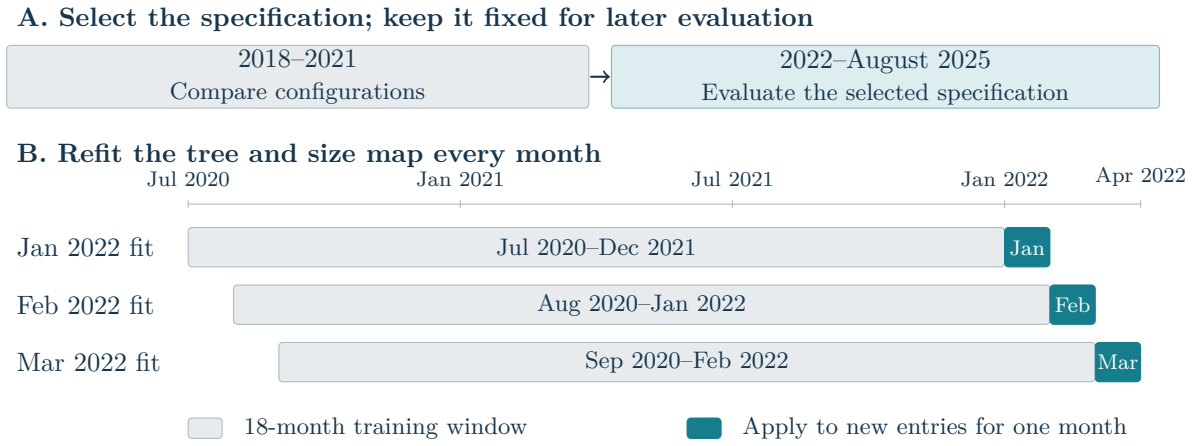


Figure 1: Two decisions, one historical dataset. The upper panel selects the 27-input, 18-month, 50/75/100% specification using earlier performance. The lower panel illustrates monthly refitting during evaluation. A training example is included only if its signal date falls within the window and its trade outcome is completed before that month's fit.

We retain the configuration with the highest annualized daily excess Sharpe on 2018–2021 among 33 existing configurations spanning training-history, feature and sizing-action choices. The selected configuration uses 27 inputs, an 18-month window and 50/75/100% actions. Candidate identifiers break score ties; its selection score is 0.575 in a $345 million research account. This configuration is then replayed at $10 million; the account assumptions are summarized in Appendix A.

Each monthly fit uses candidates that passed the entry screens on their signal dates, have valid simulated outcomes, and finished before fitting. These examples need not have been accepted by the funded account. Only signal dates within the preceding 18 months enter the fit. Tree fitting and CEM use this same sample; the wider candidate history supplies the profitability screens.

The later comparison runs from 3 January 2022 to 29 August 2025. The model specification stays fixed while monthly refitting continues: a completed March trade can inform an April fit, but an unfinished outcome cannot. New signals stop on 30 June 2025. All accounts continue from 2018 with their own accumulated cash and holdings; the 2022 boundary is a reporting split, not an account reset.

Later history had already been examined during strategy development. Monthly fits respect time order, but this period is not independent of earlier choices of contracts, screens and universe. Training also uses $0.65 commission per option contract, compared with $0.03 for every policy in final evaluation. We therefore interpret the comparison as historical evidence.

We report annualized Sharpe of daily returns above the cash benchmark, profit above cash and cash-relative drawdown. Appendix B defines these metrics and the opening portfolios at the reporting boundary.

3.4 Learning an entry size with direct policy-search RL

We use direct policy-search reinforcement learning to learn a sizing policy from historical rewards. The opportunity rule fixes the pair, direction and contracts; the policy chooses 50%, 75% or 100% of a reference budget of $450 vega per IV point on each leg, scaled by current account equity relative to its initial equity. Opportunities differ in their historical outcomes, so the learner groups them and assigns a common size to each group (Figure 2).

Each monthly regression tree receives 27 inputs: 15 numerical characteristics and 12 availability flags, covering relative IV, trailing realized variance, relatedness, maturity, the CBOE Volatility Index (VIX), costs and completed profitability. Appendix A.1 defines the inputs. It learns splits using historical full-size profit above cash as its target, splitting to reduce squared prediction error [3]. Each of its up to eight leaves is a state. The cross-entropy method (CEM) [2] then chooses one size for each state.

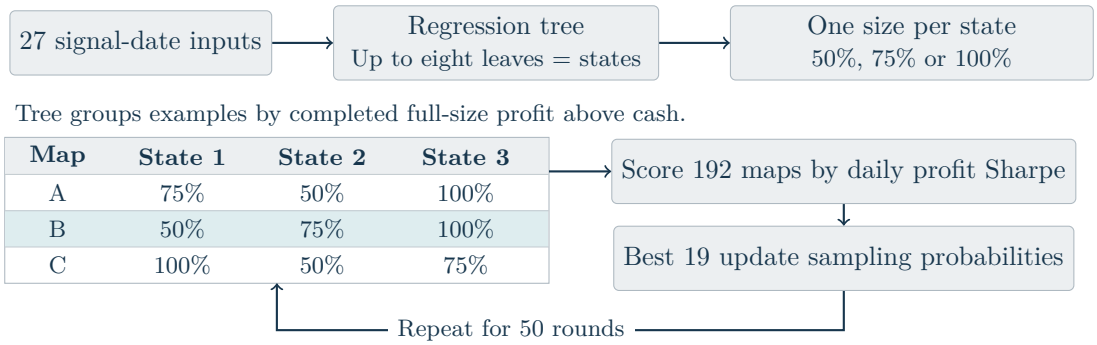


Figure 2: Tree partitioning followed by CEM sizing search. The three-state assignments illustrate candidate maps; actual trees have up to eight states. The retained map applies to the next month's new proposals.

How CEM scores a size map

A map $\boldsymbol{m}$ assigns size $a_\ell \in \{0.50, 0.75, 1.00\}$ to tree state ℓ. For completed example i, x_i is its input vector and $\ell(x_i)$ its state. Let $p_{i,d}(a)$ be profit above cash on trading day d at size a. Each size uses its own rounded contracts and standalone cash and hedge flows, rather than scaling the full-size outcome. A common stored normalization cancels from the score. Sum paths and calculate

$$g_d(\boldsymbol{m}) = \sum_i p_{i,d}\Big(a_{\ell(x_i)}\Big), \qquad \widehat{S}_{\text{train}}(\boldsymbol{m}) = \sqrt{252}\,\frac{\overline{g(\boldsymbol{m})}}{s_{g(\boldsymbol{m})}}.$$

Here g_d is combined daily profit; the numerator and denominator are its sample mean and standard deviation. The factor $\sqrt{252}$ annualizes using 252 trading days. The calendar retains overlapping trades and zero-profit sessions. This standalone-profit training score differs from funded-account Sharpe based on percentage returns and joint admission.

CEM optimizes the policy directly against the Sharpe reward, rather than estimating Q-values as in the motivating paper. It updates size probabilities toward the best 19 of 192 maps, repeating for 50 rounds. Constant maps and the best map found so far are retained as anchors. A new proposal receives its tree state's requested size, subject to account admission. The search does not establish a global optimum.

4 Benchmark and learned-sizing results

4.1 Does learned entry sizing improve the benchmark?

All policies receive the same proposals, directions and contracts. Figure 3 shows profit above cash; most of the learner's lead develops in 2022, while both lose ground during 2024.

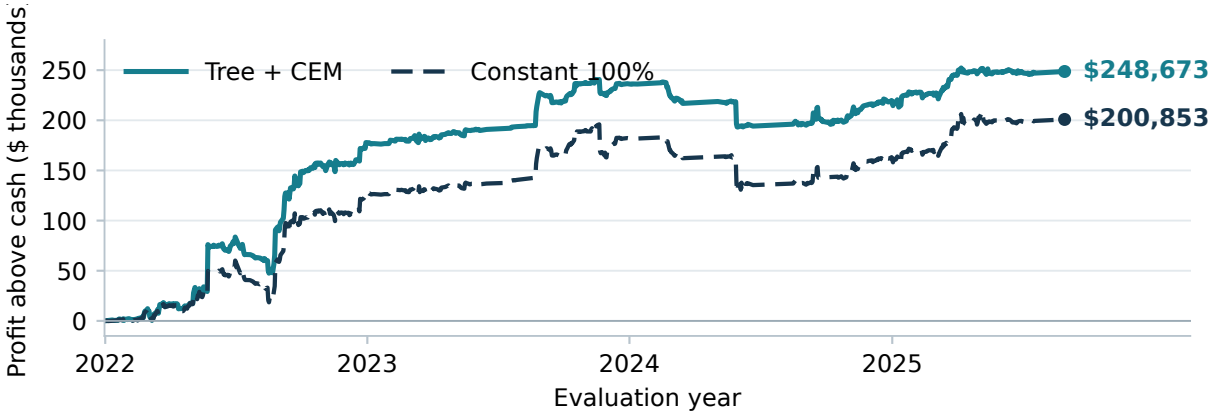


Figure 3: Cumulative profit above cash, January 2022–August 2025, rebased at 31 December 2021. Accounts retain carried positions. Endpoints match Table 2.

RL raises later Sharpe by 0.208 and profit above cash by approximately $47,800, admitting 110 entries versus 75 at full sizing. Follow-up checks test rank and regression sizing and alternative linear and additive RL policies with the same training Sharpe objective.

Policy	Earlier Sharpe	Later Sharpe	Later entries	Profit above cash ($)
Constant 50%	0.437	1.007	151	204,913
Constant 75%	0.449	0.944	104	195,021
Constant 100%	0.478	1.055	75	200,853
RL: tree + CEM	**0.573**	**1.263**	110	248,673
Spread-rank sizing	0.464	0.815	121	164,770
Ridge-score sizing	0.421	0.890	94	179,745
Direct linear policy	0.267	0.873	96	171,107
Direct additive policy	0.302	1.013	97	179,598

Table 2: Earlier: 2018–2021 replay at $10 million; later: January 2022–August 2025. The lower four rows are follow-up checks, not original selection candidates. Earlier scores are $10 million replays, not configuration-selection scores.

Tree/CEM has the highest funded Sharpe in both periods among these policies. The alternatives change the sizing pipeline, not just tree architecture. A paired circular-block bootstrap (126-session blocks; 1,000 replications) gives $[-0.103, 0.424]$ for its later gain over full sizing. This includes zero and omits research-search adjustment.

4.2 Why does the admitted portfolio matter?

At evaluation costs, summing all proposed standalone trade paths gives later Sharpe 0.606 for tree/CEM and 0.726 for the additive policy. Restricting each to its own funded admissions reverses the ranking: 1.268 versus 1.027, close to funded results of 1.263 versus 1.013. Smaller sizes can pass pair stress limits and leave room for later entries. This diagnostic indicates an interaction between sizing and admission; it neither isolates predictive sizing nor proves that the learner anticipated those constraints.

The learner's annualized total account return is 4.667%; growth relative to cash is 0.530%. Average reserve is 2.44% and peak reserve 15.77%. Cash-relative drawdown is 0.495%, compared with 0.646% at full sizing. Most total growth comes from interest, so trading value is measured above cash.

4.3 Variation across years and pairs

Table 3 shows uneven annual performance. In 2024, positive total account return accompanies a loss relative to cash.

Year	Account return (%)	Above cash (pp)	Excess Sharpe	Entries
2022	3.470	1.673	2.458	66
2023	5.288	0.432	1.492	19
2024	5.104	−0.283	−0.907	11
2025 (partial)	3.207	0.169	1.092	14

Table 3: Tree/CEM annual funded results; 2025 ends on 29 August. Account and above-cash returns are unannualized for each displayed period, including partial 2025. Above-cash figures are return differences in percentage points; Sharpe is annualized.

SNAP/UBER, GAP/TJX and AMD/MU contribute 66.9% of later net pair trading profit. The 2022 share is 83.1%. These percentages exclude shared financing; they show concentration, rather than broad repeated success across pairs.

5 Conclusion and extensions

The retained tree/CEM RL policy improves the historical funded point estimate over constant exposure and the tested alternative sizing pipelines. Its benefit combines exposure choices with portfolio admission. The uncertainty interval, previously inspected history, inherited universe and simplified execution limit conclusions about repeatability and live profitability.

A natural extension is allocation across several independently researched signals. Freeze each signal's trade proposals, then let RL choose exposures using current inventory, available capital, shared underlying risks and completed signal performance. Train against the actual funded ledger and compare with fixed allocations. This would test whether learning can combine opportunities into a portfolio, beyond sizing one retained signal.

A Signal and learning specification

Choose the expiry closest to 30 days in the historical anchor selection, then the put in that expiry nearest absolute delta 0.25, within 0.075. Ties favor narrower spreads, higher open interest and lower contract identifier. Entry requires standard 100-share contracts, positive noncrossed quotes, open interest at least 100, width at most 25% of midpoint and 7–55 days remaining. The IV rank includes the signal close; relatedness is fixed monthly from the prior 126 sessions. Profitability memory uses completion dates, not entry dates.

Admission stress sums each name's worst modeled option-plus-hedge loss across spot shocks of $-20/0/+20\%$ and IV shocks of $-10/0/+10$ percentage points, without credit for pair correlation. Stress loss is capped at 5% of budget; reserve (gross option premium plus half netted hedge notional) at 80%, hedge notional at twice budget and gross vega at \$20,000 per IV point per \$1 million. Global checks use account equity; pair checks use \$1 million scaled by current/initial account equity. Proposals are processed in pair-identifier order. These are research limits, not broker margin. Corporate events or unsupported declared distributions exit affected pairs; missing entry quotes reject proposals, while held marks may use last-known IV with current spot, rates and known dividends. Expiry is cash-settled with a European valuation approximation; market impact, early assignment and forced liquidation are not fully modeled.

The tree has up to eight leaves, with minimum leaf count one eighth of its training sample rounded down, bounded between 8 and 63. Fewer than 16 examples use full sizing. CEM begins uniformly each month and mixes 30% current probability with 70% elite frequency, multiplies by 0.97 and adds 0.03 equally across the three actions. The random seed is 2026.

A.1 Inputs available at the signal date

The model uses the following 15 numerical inputs. Proposed-direction quantities are oriented as long leg minus short leg. IV and volatility levels are decimal annualized values. Trailing realized variance is 252 times mean squared daily simple returns over the latest 20 listed trading sessions, with at least 10 observations, including the signal-close return. Returns generally exclude dividends; adjusted-close percentage changes provide the fallback series.

Input	Definition
Mean IV	Mean of the two selected-contract IVs.
Oriented IV gap	Long-minus-short IV divided by mean IV.
Rank strength	Twice the absolute distance of the IV-gap percentile from 0.5.
Oriented variance-premium gap	Long-minus-short difference in IV squared minus trailing realized variance, divided by mean IV squared.
Mean RV	Mean of the square roots of the two trailing realized variances.
Beta; correlation	Two inputs: 126-session A-on-B beta and correlation, fixed at monthly formation.
Mean expiry; expiry gap	Two inputs: mean calendar days to expiry and long-minus-short expiry days.
Commission in IV points	Sum across legs of \$0.65 divided by each contract's dollar vega per IV point.
VIX	Most recent strictly prior VIX close divided by 100.
91-day profit; count	Two inputs: completed full-size profit per initial vega averaged over the trailing 91 calendar days and the number of completed outcomes. The profit input is divided by 100 as a fixed scale.
365-day profit; count	Two inputs: the same measurements over 365 calendar days.

The first three inputs are core fields. Each of the other 12 has a binary availability flag, giving 27 inputs in total. Nonfinite optional values receive zero and flag zero; observed values have flag one. Profit features use only outcomes completed before the signal date; future returns and profits enter outcomes, not inputs.

B Performance definitions and evidence

Let N_t be account equity and K_t all-cash benchmark wealth at close t. Both start at \$10 million in January 2018. The benchmark compounds the previous available WRDS zero-curve rate nearest 30 days over calendar days using a 365-day year. Daily excess return and annualized Sharpe are

$$e_t = \left(\frac{N_t}{N_{t-1}} - 1\right) - \left(\frac{K_t}{K_{t-1}} - 1\right), \qquad \widehat{S} = \sqrt{252}\,\bar{e}/s_e.$$

Here $\bar{e}$ and s_e are the sample mean and standard deviation of daily excess returns; 252 is assumed trading days per year. For a reporting period with preceding close 0 and final close T, dollar profit above cash is $(N_T - K_T) - (N_0 - K_0)$. Set $Q_t = N_t/K_t$: annualized relative growth is $(Q_T/Q_0)^{1/Y} - 1$, where Y is elapsed days divided by 365.25. Cash-relative drawdown is the largest percentage decline in Q_t from its running peak within that period, including close 0.

Annual-table account return is $N_T/N_0 - 1$; above-cash percentage points subtract benchmark return $K_T/K_0 - 1$. Period returns are unannualized, including partial 2025; Sharpe remains annualized. At the 31 December 2021 boundary, learner and constant-full equity are \$10.568 million and \$10.581 million, with four and three open trades respectively. Benchmark wealth is \$10.510 million. Chart curves subtract each policy's opening dollar excess wealth; accounts and holdings are not reset.

The gain interval resamples the learner and constant-full daily excess returns with identical indices: 1,000 samples concatenate randomly started 126-session blocks, wrap at the series end and truncate to original length. Each sample recalculates the Sharpe difference; endpoints are its 2.5th and 97.5th percentiles. Fitted policies are fixed and research selection is unadjusted.

Evidence sources. Results and implementation use repository snapshot 688bc39. The directory `reports/p25_validation_split` contains `periods.csv`, `annual.csv`, `holdout_intervals.csv` and `selection.json`. Figure 3 uses the selected account in `data/p25_validation_split_experiment.zip` and the constant-full account under `data/submission/artifacts/contracts/p25_capital_sensitivity`. The complete 125-pair roster, candidate-date ranges and proposal counts are retained in the companion `paper/pair_universe.csv`; ranges describe recorded candidates, not continuous quote coverage or executed trading dates. Inherited formation calendars and usable contracts condition the universe. Supporting comparisons and concentration use `reports/p25_strategy_audit` and `reports/p25_direct_policy`; the admission diagnostic is in `reports/p25_generalization_bridge`. Vendor data retain their institutional license restrictions.

References

[1] B. Ning and K. Lee (2024). *Advanced Statistical Arbitrage with Reinforcement Learning*. arXiv:2403.12180v1. https://arxiv.org/abs/2403.12180.

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FOLLOW-UP EXTENSION · 15 OCTOBER 2026

Extension 1. Adding a second volatility signal

The preceding paper studies one economic signal: relative implied volatility in approximately 30-day, 25-delta put pairs. Its conclusion proposes allocating across several signals. This follow-up implements a first version of that extension while retaining the original signal, actual-contract simulator and entry-sizing result.

Persistent realized-versus-implied differences

The second signal asks whether a name's recent options have repeatedly implied more volatility than subsequently realized. For each name, we average completed 30-day realized volatility minus entry ATM call IV over the preceding year. Only outcomes finished before the decision enter this memory. We rank eligible names daily and sell the bottom decile: the names with the most persistent negative RV-minus-IV history. This simple persistence forecast avoids changing the original pair signal.

We express the signal through approximately 30-day ATM straddles: equal-count short calls and puts with the same underlying, strike, expiry and adjustment. The combined reference vega is $450 per IV point. Delta is hedged daily and contracts normally run to expiry. The original call-IV input is retained when moving from calls to straddles. The simulator applies the same funded cash-flow accounting, evaluation execution charges, dividends, financing and admission framework as before.

Learning size within the second signal

A monthly ridge score partitions proposals into three contexts using 60 months of completed history. The inputs include IV, completed RV-minus-IV memory, VIX, recent IV/VIX changes and a causal fixed-half benchmark's trailing performance and capital use. Exhaustive search over 27 maps assigns 50%, 75% or 100% size to each context to maximize daily training Sharpe. All 35,774 proposals remain; early months use half size until enough completed examples exist.

Straddle entry policy	2018–21 Sharpe	2022–25 Sharpe	Later entries	Later profit
Fixed 50%	0.740	0.920	1782	$1.824m
Fixed 75%	0.542	0.968	1222	$2.010m
Fixed 100%	0.528	0.990	940	$2.206m
Contextual sizing	0.775	1.041	1204	$2.379m

The retained bandit improves the historical point estimate against the three fixed sizes. Straddles also generate substantially more dollar profit and return volatility than the original put-pair strategy. This difference creates an allocation problem: how selectively should the portfolio take the larger short-volatility exposure?

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Extension 2. Interpreting entry-size decisions

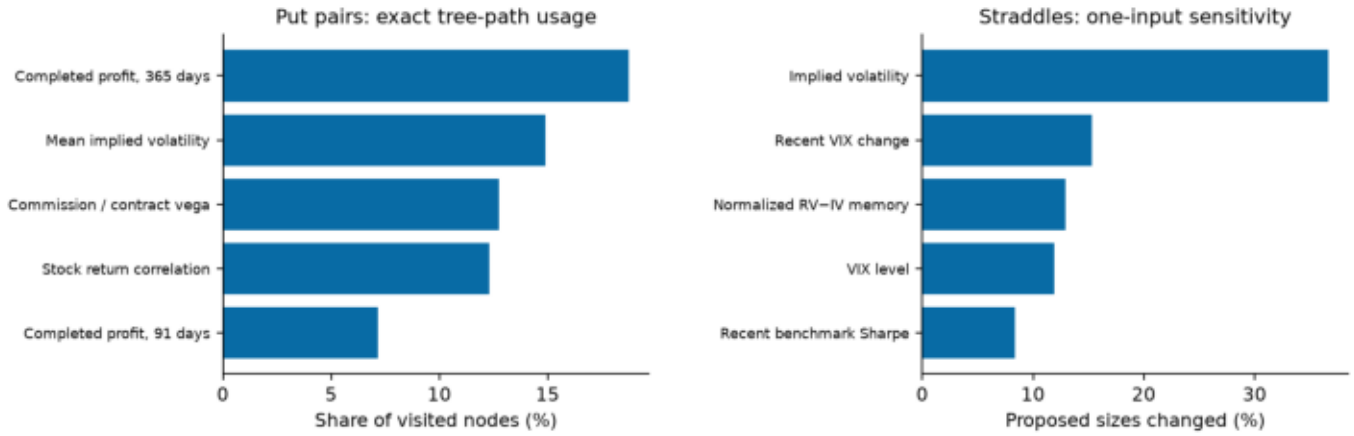


Figure E1. Entry-policy explanations. Put bars measure actual tree-path usage; straddle bars measure how often centering an input changes proposed size. The measures have different definitions and are not causal effects.

The original put learner can be explained by following its exact tree path and reading the size assigned to the resulting leaf. Completed 365-day profit appears in 18.7% of later visited decision nodes; mean IV, commission relative to vega and stock-return correlation account for 14.9%, 12.7% and 12.3%. These frequencies describe what the model uses, not whether each feature causes profit.

For straddles, standardized ridge contributions add exactly to a context score; thresholds and the saved context-to-size map then determine exposure. Centering IV changes 36.6% of fitted later proposal sizes. Recent VIX change changes 15.3%, normalized RV-minus-IV memory 12.9% and VIX level 11.9%. The action map need not be monotonic in the score because it optimizes aggregate Sharpe. All 608 put-pair sizes and 35,774 straddle sizes reproduce the frozen policies exactly.

Later entries versus fixed full	Both enter	Bandit only	Fixed only
Original put-pair strategy	74	36	1
New straddle strategy	897	307	43

The entry benefit combines exposure choices and admission, extending the original paper’s capacity diagnostic. Smaller commitments can leave room for later trades. The put bandit admits 110 later entries versus 75 at full size; the straddle bandit admits 1,204 versus 940. These are complete funded policies, rather than size comparisons on identical executed portfolios.

There is also descriptive outcome separation. Among quote-available completed straddle labels, full-size profit averages $1,631 for later proposals assigned 100%, versus $16 for those assigned 50%. This subset is conditioned on available historical labels. Three within-date size shuffles reach later Sharpes of 0.834, 0.928 and 1.044, versus the bandit’s 1.041; one control wins. We therefore separate decision transparency and funded performance from stronger claims of predictive sizing.

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Extension 3. Learning the capital split



Figure E2. The extension adds a second signal and an outer allocation decision. The original put-pair framework remains the first strategy; its retained entry policy is frozen.

We freeze the retained put-pair and straddle entry policies and let an outer Q learner allocate one $10 million budget between their funded strategy indices. Actions choose put/straddle weights of 100/0, 90/10, 75/25, 50/50, 25/75, 10/90 or 0/100. Weights sum to one. All seven fixed benchmarks use the same menu and start at their own split. There is no outer leverage, cash action or additional portfolio stop.

State and execution

The state combines put-spread strength, the put implied-versus-realized variance-spread feature, straddle IV-versus-RV richness, completed straddle premium, relative strategy volatility, strategy return correlation, VIX, relative recent Sharpe and previous straddle weight. Market information uses the prior close. A weekly decision takes effect at the next close, leaving the first session at old weights. Holdings then drift and affect the cost of the next resize.

The action-value function is quadratic in straddle weight, with coefficients linear in standardized state inputs. Monthly fitted Q iteration uses the preceding 104 completed weeks, requiring 52. The seven prior allocations crossed with seven actions are counterfactuals on the same week; they do not create independent observations. Q uses 50/50 before enough history is available, while fixed controls always use their own target.

Reward and sequential value

Weekly reward sums daily excess returns normalized by completed training-window reference volatility, then subtracts λ times squared normalized returns. λ controls risk preference. The separate discount $\gamma = 0.99$ weights continuation value. Ridge regularization and 200 Bellman updates are held fixed. A matched $\gamma = 0$ learner provides a one-period contextual-allocation control.

This is fractional strategy-index allocation, preserving source returns and adding source-based resizing charges. It is not a shared integer-contract replay across both signals. The straddle sleeve introduces net short vega, so the combined portfolio is not vega neutral. The extension tests capital allocation across the existing strategies, while retaining the original contract-level study as its foundation.

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Extension 4. Fixed versus dynamic allocation

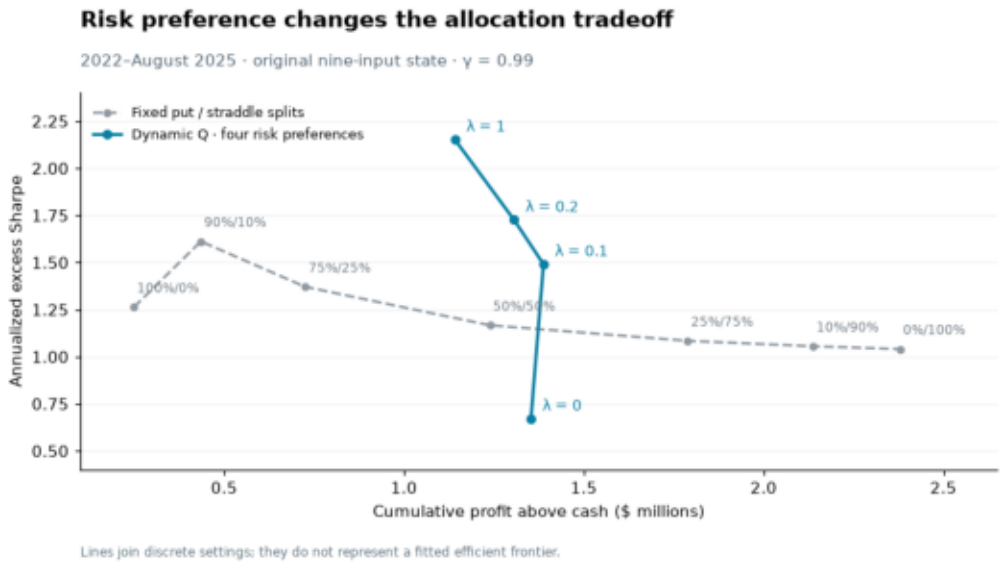


Figure E3. Four risk preferences with the original nine-input Q state versus seven fixed capital splits, 2022–August 2025. Gray labels give put/straddle weights. Lines connect discrete results rather than a fitted frontier; the table also reports validation performance.

Risk penalty	2018–21 Sharpe	2018–21 profit	2022–25 Sharpe	2022–25 profit
λ = 0	1.427	$1.281m	0.671	$1.353m
λ = 0.1	1.286	$1.003m	1.491	$1.388m
λ = 0.2	1.215	$0.875m	1.726	$1.306m
λ = 1	1.005	$0.660m	2.152	$1.141m

We report 2018–2021 validation and 2022–August 2025 walk-forward out-of-sample performance. Each monthly fit uses only outcomes completed before its decisions; earlier completed evaluation-period weeks can therefore enter later refits. The four settings illustrate risk preferences; they are not all validation winners. Features, source bandits, allocation choices, execution and discount remain identical.

At penalty 0.20, Q reaches later Sharpe 1.726 and $1.306 million profit above cash. The highest fixed-split Sharpe is 1.612 at 90/10, earning $435,181. Penalty 0.10 retains more profit at Sharpe 1.491. Raising the penalty to 1.00 reaches Sharpe 2.152 while reducing profit to $1.141 million. Straddle-heavy fixed mixes still earn more dollars: these are different risk–return tradeoffs, rather than dominance on every objective.

The zero-penalty policy provides a useful contrast. Its validation Sharpe is 1.427, but later Sharpe falls to 0.671. Without a variance penalty, the reward favors profit and the policy allocates more often to the higher-profit straddle signal. A strong validation Sharpe alone does not ensure that this risk preference transfers to a different period.

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Extension 5. Why the switches help

Subsequent straddle performance	Mean/day (bps)	Daily vol. (bps)	Conditional Sharpe
Q chose straddles	3.05	16.18	3.00
Q chose put pairs	1.30	27.00	0.76
Q chose a mixed split	-2.33	20.00	-1.85

The allocator's effective distinction is when straddle carry compensates for its risk. After Q chooses straddles, their average excess return is higher and daily fluctuations lower than on dates when Q chooses put pairs. On the latter dates, the put strategy has conditional Sharpe 1.78. The allocator can therefore improve Sharpe while forgoing some average straddle profit.

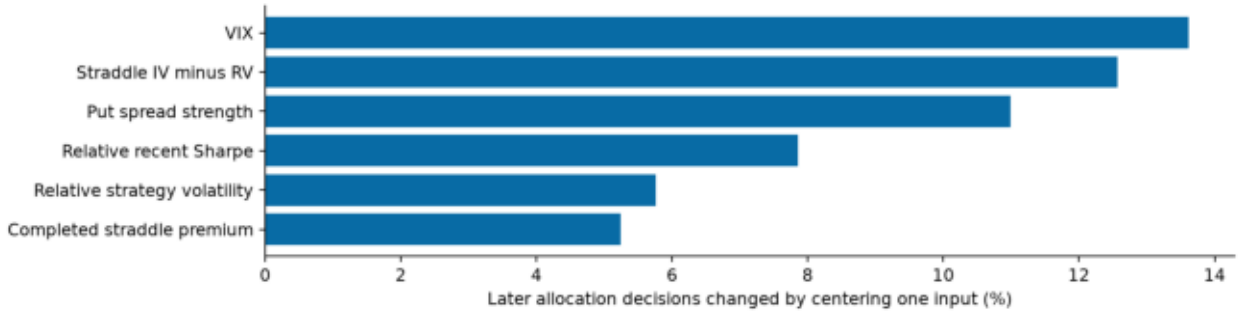


Figure E4. Penalty-0.20 Q: fraction of later decisions changed when one input is set to its monthly training center, holding the other actual state values fixed.

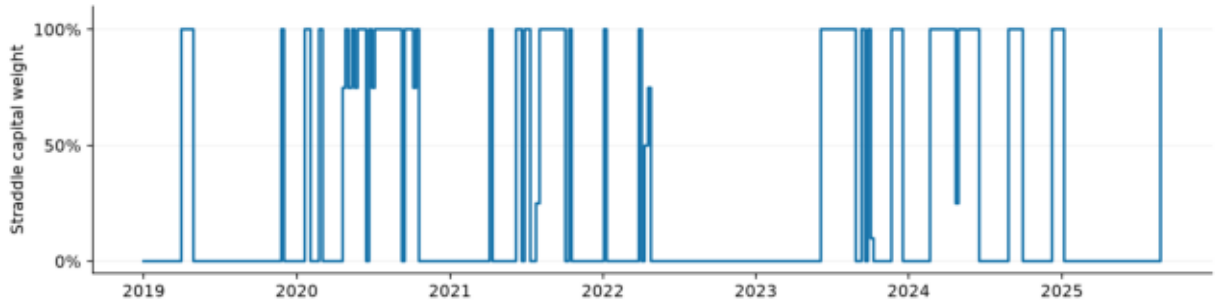


Figure E5. The saved Q policy mostly switches between the two strategy indices. These are capital weights, rather than the fraction of cash spent on options. Only fitted decisions are shown.

Of 191 later decisions, 141 choose put pairs alone and 46 straddles alone; only four choose interior splits. Mean straddle target is 24.9%. In later low-VIX weeks it averages 44.9%, versus 4.7% in high-VIX weeks. This median split describes observed behavior, not a fitted threshold rule; the earlier relationship differs. The inputs jointly influence decisions through monthly refits.

The table uses source returns strictly after the decision's execution close through the next execution close. Source costs are included; additional allocation-resizing charges are excluded. These are conditional statistics on noncontiguous historical dates, not separately deployable strategy Sharpes. Centering inputs is a policy sensitivity check, not a causal feature test.

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Extension 6. Timing, continuation and interpretation

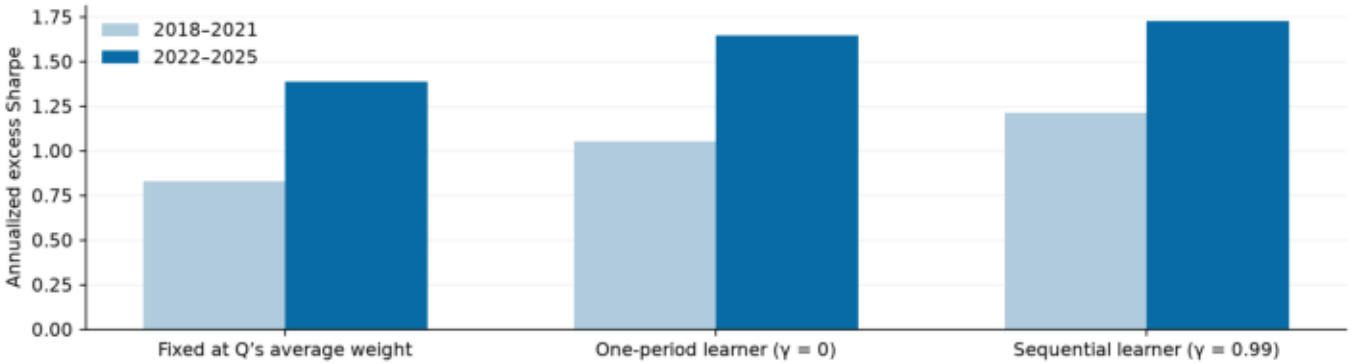


Figure E6. Matched controls at penalty 0.20. The average-weight control uses the realized mean target retrospectively; the discount-zero learner uses the same reward, features, data and regularization as Q.

Penalty-0.20 policy or control	2018–21 Sharpe	2022–25 Sharpe	Later profit
Fixed at Q's mean target	0.828	1.387	$0.841m
One-period learner, γ = 0	1.053	1.645	$1.184m
Sequential Q, γ = 0.99	1.215	1.726	$1.306m

A constant target equal to Q's later mean straddle allocation gives Sharpe 1.387 and $841,251 profit, versus Q's 1.726 and $1.306 million. The earlier comparison is 0.828 versus 1.215. Average exposure alone therefore does not explain the point estimate. These targets use the realized mean retrospectively and are diagnostic rather than validation-selected policies.

Shifting the learned action schedule through 49 nonzero circular offsets, preserving its action counts and most serial structure, gives median later Sharpe 0.744; none reaches Q. Three of 49 earlier rotations reach or exceed the actual policy. Wealth and resizing costs are recomputed. This is descriptive evidence for timing conditional on the researched schedule, not a selection-adjusted significance result.

The matched one-period learner already reaches later Sharpe 1.645. Q's continuation term changes 9 of 191 later decisions when evaluated at the same actual states; it adds a smaller increment than the full dynamic-versus-fixed gap. Some feature-centering interventions also improve Sharpe slightly. The clearest interpretation is conditional allocation with a modest incremental sequential benefit, rather than proof that every input or long-horizon planning is essential.

The reward changes behavior, not just the reported score. With no variance penalty, 138 of 191 later decisions choose straddles alone, versus 46 at penalty 0.20 and 31 at 1.00. Mean straddle targets are 73.2%, 24.9% and 16.9%. The zero-penalty policy loses $331,255 above cash in 2025; penalty 1.00 earns $75,335. Fixed 75% straddles still achieves later Sharpe 1.084, so timing as well as exposure contributes to the zero-penalty failure.

Increasing Bellman updates from 200 to 500, 1,000 and 2,000 leaves every later allocation decision and result unchanged for all four settings. The weak zero-penalty result is therefore not explained by an insufficient iteration budget in this check.

The extension separates signal profitability from capital allocation. Modest positive penalties retain strong Sharpe across both periods, while the profit-only objective transfers less reliably. Shared integer-contract replay and execution sensitivity are the next checks.